



Advanced Data Analytics (ADA)

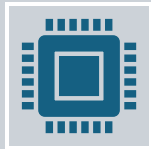
Module 4

Synthetic Data Generation

Introduction to Synthetic Data Generation



What is Synthetic Data? Artificially created data that mimics real-world datasets.



Why it matters: Used for privacy protection, data augmentation, and simulation.



Goal: Learn to generate and evaluate realistic synthetic data for analytics workflows.

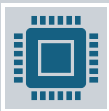
Why Generate Synthetic Data?



Data Scarcity: Fill gaps when real data is limited.



Privacy & Compliance: Avoid exposing sensitive data General Data Protection Regulation and Health Insurance Portability and Accountability Act (GDPR, HIPAA).



Simulations & Edge Cases: Model rare or future scenarios.



Augmentation: Enhance datasets to improve ML model performance.

Classical Methods of Data Synthesis

- Random Sampling: From known distributions.
- Bootstrapping: Resampling with replacement.
- Noise Injection: Add realistic variability.
- Rule/Template-Based: Generate logical, domain-relevant data (e.g., using Faker).

Comparison of Classical Methods of Data Synthesis

Method	Description	When to Use	Example
Random Sampling	Generate values from predefined statistical distributions (e.g., Normal, Exponential).	When variable distributions are known and independent.	Simulate customer ages using a Normal($\mu=35$, $\sigma=10$).
Bootstrapping	Create new samples by resampling existing data with replacement.	When you have a small dataset but want more samples while preserving structure.	Resample 1000 records from 100 observations of loan data.
Noise Injection	Add small random variations (noise) to real values to create realistic diversity.	When simulating measurement errors or small natural variations.	Add Gaussian noise to income or sensor readings.
Rule / Template-Based	Use logic or templates to generate structured, domain-valid data.	When data must follow business rules or realistic relationships.	Use Faker to create names, emails, and consistent credit policies.

Probabilistic & Statistical Modeling

Goal: Preserve relationships between features.



Techniques:

- Copulas: Model dependencies separately from distributions.
- Bayesian Models: Represent uncertainty and conditional dependencies.
- PGMs: Simulate data respecting dependency structures.

Comparison between statistical synthetic data generation model

Method	Description	When to Use	Strengths
Copulas	Mathematical functions that model dependencies separately from the marginal distributions of each variable.	When you need to preserve correlation and dependency structures between multiple variables.	Captures realistic relationships between variables; flexible for mixed data distributions.
Bayesian Models	Represent uncertainty by treating parameters and variables as probability distributions rather than fixed values.	When you have limited data, need interpretability, or want to incorporate expert knowledge.	Handles uncertainty well; transparent; integrates prior knowledge; generates explainable synthetic data.
Probabilistic Graphical Models (PGMs)	Graph-based models (e.g., Bayesian Networks) that represent and simulate dependencies among variables.	When you need to generate synthetic data that respects known conditional dependencies or causal structures.	Excellent for complex dependency modeling; visually interpretable; supports conditional reasoning.

Machine Learning-Based Generation

When to Use: Complex, high-dimensional, mixed data.

Advantages:

- Learns non-linear relationships.
- Handles mixed data types.
- Reduces manual modeling effort.

Tools: GANs, VAEs, and SDV library.

Introduction to SDV (Synthetic Data Vault)

Supports: Tabular, relational, and time series data.

Models:

- CTGAN (GAN-based) Conditional Tabular Generative Adversarial Network
- TVAE (VAE-based) Variational Autoencoder
- GaussianCopula (Statistical): A **Gaussian Copula** assumes that the **relationships** (dependencies) between variables can be represented as if they come from a multivariate normal distribution even if each individual variable has a different distribution shape.

Workflow: Fit model → Generate data → Evaluate results.

Comparison between SDV common models

Model	Full Name / Type	Description	Strengths
CTGAN	Conditional Tabular Generative Adversarial Network (GAN-based)	Uses two networks (Generator & Discriminator) to generate realistic tabular data by conditioning on categorical variables. Handles imbalanced and mixed-type data well.	Captures complex, nonlinear relationships; works well with categorical + numerical data; handles class imbalance.
TVAE	Tabular Variational Autoencoder (VAE-based)	Learns latent representations of tabular data using an encoder-decoder architecture to reconstruct and generate realistic samples.	Good for continuous and correlated features; stable training; interpretable latent space.
GaussianCopula	Statistical Model	Models the dependencies between variables using a Gaussian (normal) copula to preserve correlation structures.	Fast, interpretable, and preserves feature correlations; ideal for structured numeric data.



Evaluation of Synthetic Data

Why Evaluate: to ensure realism, utility, and privacy.



Dimensions:

1. Statistical Similarity: Distribution, correlation.
2. Utility: TSTR (Train on Synthetic, Test on Real).
3. Privacy: Check for duplicates or near matches.

Tools for Evaluation

- sdmetrics- Synthetic Data Metrics Library : an open source library that lets you evaluates statistical similarity, ML utility, and privacy.



Key Metrics:

- KS test for distributions.
- Correlation preservation.
- Privacy leakage detection.